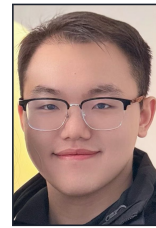


Zhiyuan Ouyang

2nd-year Ph.D. @ East China Normal University / Shanghai Institute of Intelligence

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Summary

My research centers on **Generative Models**, focusing on Diffusion / Flow-Matching architectures and underlying topologies to enhance generation efficiency and controllability. I possess **full-stack edge AI delivery capabilities**, independently handling core algorithm design, model training, and native edge deployment. In AI engineering, I have **large-scale distributed training** expertise, familiar with Slurm and multi-node DDP cluster scheduling, with hands-on experience in concurrent training and communication tuning. Furthermore, I specialize in **custom inference infrastructure**, excelling in inference acceleration and extreme VRAM optimization using TensorRT and ONNX Runtime.

Education

ECNU/SII 2nd-year Ph.D. in Computational Mathematics

Shanghai, China

Advisor: Prof. Junchi Yan

Honors & Awards: National Scholarship for PhD Students, China (Top 1.5%)

Competition Awards: C/C++ Programming Contest, Collegiate Division (Provincial First Prize, Top 5%)

Research Interests: Generative Modeling Theory & Multi-modal Generation / Time-Series Analysis

Publications

Zhiyuan Ouyang (First Author). Primal-Spectral Generative Modeling: Fast Analytical Generation via Pseudoinverse Lévy Inversion. *Published in ICML 2026 (CCF-A)*

Zhiyuan Ouyang (First Author). A Novel Network for Resolving Subjective Masking Differences and Accurate Thyroid Nodule Diagnosis. *Published in Computational Biology and Chemistry (JCR-Q2)*

Zhiyuan Ouyang (First Author). Model Detection for Grey Forecasting Model with Polynomial Term. *Published in Communications in Statistics (JCR-Q3)*

Projects

Generative Modeling in Characteristic Function Space (Outstanding Ph.D. Project) 📄 **PaperNov 2025 – May 2026**

- **Theoretical Innovation:** Explored the evolution mechanism from Primal Data Distribution to Characteristic Function. Transformed the distribution into a continuous function via Spectral Methods, solving the challenge of neural networks reconstructing global topology from sparse samples, and providing theoretical guarantees for convergence.
- **Algorithm Architecture:** Proposed the **PriSpecNet** architecture and designed a single-step evaluation (1-NFE) Pseudoinverse Lévy Inversion (PiLI) solver. This transforms the generation process into a Closed-form Analytic problem, completely eliminating the reliance on iterative Integration while maintaining full compatibility with Stochastic Interpolation.
- **Application Performance:** Unified generation and forecasting in the time series domain, reducing Context-FID metrics on Sines, Solar, ETTh, and Stock datasets by 50.0%, 41.5%, 80.6%, and 63.1%, respectively. In the ImageNet 256×256 generation task, achieved an FID of 1.66 with only **26 Gflops** computational overhead, realizing a **170x** computation reduction compared to 25-NFE DPM-Solver++.

Controllable Video Generation Project (University-Industry Collaboration, Project Lead) Oct 2025 – Jun 2026

- **Project Leadership:** Served as project lead coordinating university and industry requirements, driving requirement decomposition, technical decisions, and milestone delivery; investigated representative approaches including **Control-A-Video** and **ControlVideo**, defining a **Video ControlNet**-centric solution.
- **Generation Pipeline & Data Synthesis:** Led the implementation of a controllable video generation workflow based on **ComfyUI** and **Stable Diffusion**. Built automated multimodal data synthesis pipelines utilizing VLMs for offline annotation and distillation, significantly improving motion following and visual consistency.
- **Inference Acceleration & Edge Delivery:** Addressed high latency and VRAM bottlenecks in video generation by rebuilding custom inference engines using **TensorRT / ONNX**, achieving KV-Cache optimization and cross-frame

computation reuse; accomplished native edge deployment for stable on-premises delivery.

Internship

Baizhi Xunlian Technology Co., Ltd. (Head of AI Department)

Sep 2023 – Sep 2024

- **Generative Model R&D:** Led the construction of visual generation Pipelines based on **ComfyUI** and **Stable Diffusion**. Introduced SFT and human preference alignment (RL) for vertical domain Fine-tuning, delivering highly controllable, End-to-End customized image generation solutions.
- **Custom Infrastructure Deployment:** Responsible for architecture planning of enterprise AI production environments. Conducted **Inference Optimization** for heterogeneous platforms, refactoring critical inference operators with C++/Rust to break through edge deployment bottlenecks, achieving low-latency and low-VRAM high-availability deployment.
- **Team Management:** Comprehensively managed the technical evolution and Agile R&D of the AI department. Provided AIGC commercial landing consulting; established internal engineering standards, led technical training, and drove efficient cross-functional project delivery.

Technical Skills

Programming Languages: **Python** - proficient in algorithm modeling and data exploration; **C++ / Rust** - leading high-performance edge inference engines and native deployment, adept at using AI coding assistants for efficient development.

Algorithm & Training: Deep understanding of Diffusion Models (DiT, U-Net), Flow-Matching, SFT / RL; proficient in **PyTorch** and **JAX**, with expertise in FlashAttention-2, FlexAttention, multi-node DDP, and mixed-precision training.

Inference & Architecture: Specialized in custom inference pipelines, mastering TensorRT, ONNX Runtime (CUDA / DirectML / Metal), and quantization (Q4/Q8); expert in sampler optimization (e.g., 1-NFE solvers) and operator fusion for extreme compute and VRAM compression.

Data Synthesis & Deployment: Proficient in large-scale multimodal data synthesis and VLM-based distillation; capable of edge AI loop deployment, achieving lightweight, zero-Python dependency native edge integration.